

MIDTERM (SHORT NOTES)

2024 - 2025

CS201 Short Questions

Contact Us: 03254755408

Lecture 01:

What is a program?

A precise sequence of steps to solve a particular problem.

Why is programming important?

It helps develop analytical and problem-solving skills.

What are some important programming skills?

Attention to detail, reusability, user interface design, and code commenting.

Why is attention to detail necessary in programming?

To ensure the program runs correctly and produces logical results.

What does reusability in programming mean?

Writing code that can be used again for similar tasks.

Why is user interface design important?

To make programs easy for users to understand and interact with.

What is the purpose of commenting code?

To describe the purpose and functionality of code for future reference.

Why are computers considered “stupid”?

Because they only do exactly what they're told, nothing more.

What is a program design recipe?

A structured method to analyze problems and build correct programs.

What is the role of testing in program development?

To evaluate, check, and refine the program for accuracy.

Lecture 2

What are the two main categories of software?

System software and application software.

What is the role of an operating system?

It manages hardware resources and provides services for software.

What is a device driver?

Software that allows the OS to communicate with hardware devices.

What is utility software?

Programs that perform specific system-related tasks like disk cleanup or defragmentation.

Give an example of application software.

Accounting software or a GPS navigation app.

When and where was the C language developed?

In the 1970s at Bell Laboratories.

Who developed the C language?

Dennis Ritchie.

Why was C language created?

To write system software, including the UNIX operating system.

What is a text editor used for in programming?

To write and edit the source code.

What is the function of a compiler?

It translates the entire source code into machine language.

Lecture 3

What does `#include <iostream.h>` do?

It includes the input/output stream library in the program.

What is the role of `main()` in a program?

It marks the starting point of program execution.

What does `cout` do?

Displays output on the screen.

What is a variable?

A named memory location used to store data.

What is the assignment operator in C?

The = symbol.

What are the basic data types for whole numbers in C?

int, short, and long.

Which data types are used for real numbers?

float and double.

How is a character stored in C?

Using char type and enclosed in single quotes.

What is the % operator used for?

To get the remainder in integer division (modulus).

What is operator precedence?

The rules that determine the order of operations in expressions.

Lecture 4**What does cin do in C?**

It takes input from the user.

How is user input stored in a variable?

Using `cin >> variable;`.

Why is int not ideal for storing average age with decimal?

Because it truncates the decimal part.

What is a common way to calculate an average in C?

Divide the total by the number of values.

How do you separate digits of a number in C?

Using modulus and division by 10.

What is the output of `1234 % 10`?

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What is the purpose of parentheses in expressions?

To force the order of evaluation.

Why is operator precedence important in expressions?

It affects the result of arithmetic calculations.

How can you reuse a variable for memory efficiency?

By assigning it a new value during the program.

What is the output of $x = 2 + 4 * 3$?

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Lecture 5

What is a conditional statement?

It allows decisions to be made in a program.

What is the syntax of an `if` statement?

```
if (condition) { statements }
```

What are relational operators?

Operators that compare values (e.g., `>`, `<`, `==`).

What does `==` mean in C?

Checks if two values are equal.

Why is `=` different from `==`?

`=` assigns a value; `==` checks for equality.

What is a block of statements in C?

A group of statements enclosed in `{ }`.

What is the purpose of indentation in code?

To make the code readable (no effect on execution).

What is a flow chart?

A diagrammatic way to represent a program.

What symbol represents a decision in a flow chart?

A diamond.

What kind of structure does an `if` statement represent?

Decision-making or branching structure.

Lecture 6

What is the purpose of `if-else` structure?

To execute one block if a condition is true and another if it's false.

What is a logical operator?

An operator used to combine multiple conditions.

What are common logical operators in C?

&& (AND), || (OR), ! (NOT).

What does && do in C?

Returns true if both conditions are true.

What does || do in C?

Returns true if at least one condition is true.

What does ! do in C?

It reverses the truth value of a condition.

How does nested if work?

An if statement inside another if statement.

What is the difference between = and == again?

= assigns; == compares.

What happens if a condition is false in if-else?

The else block is executed.

Why is it useful to test different inputs in a program?

To ensure the logic handles all possible cases correctly.

Lecture 7

What is the purpose of loops in programming?

To repeat a set of instructions multiple times.

What are the types of loops in C?

while, do-while, and for loops.

What is the structure of a while loop?

```
while (condition) { statements }
```

When does a while loop terminate?

When the condition becomes false.

What is a loop control variable?

A variable that controls how many times the loop runs.

What can cause an infinite loop?

A condition that never becomes false.

How can you prevent infinite loops?

By updating the loop control variable correctly.

What is the minimum number of times a `while` loop executes?

Zero, if the condition is false initially.

What does `i++` mean?

Increment variable `i` by 1.

What is a common use of loops in programs?

Reading input, computing sums, or printing sequences.

Lecture 8

What is the syntax of a `do-while` loop?

```
do { statements } while (condition);
```

When does a `do-while` loop execute its body?

At least once, regardless of the condition.

What is the key difference between `while` and `do-while`?

`while` checks the condition first; `do-while` checks after execution.

When is a `do-while` loop useful?

When the body must run at least once.

Can you use `break` in a loop?

Yes, to exit the loop immediately.

Can you use `continue` in a loop?

Yes, it skips the rest of the loop and starts the next iteration.

What does an infinite `do-while` loop look like?

```
do { } while (1);
```

How do you validate input using `do-while`?

Repeat until correct input is entered.

What kind of errors can improper loop use cause?

Infinite loops, skipped iterations, or logic errors.

Where is the semicolon placed in a `do-while` loop?

After the `while(condition);`

Lecture 9

What is a `for` loop used for?

To repeat actions a known number of times.

What is the syntax of a `for` loop?

```
for(initialization; condition; update) { statements }
```

What are the three parts of a `for` loop?

Initialization, condition, and update.

Can all loops be written using `for`?

Yes, any `while` or `do-while` can be written using `for`.

When is a `for` loop ideal?

When the number of iterations is known beforehand.

Can you nest `for` loops?

Yes, you can place one `for` loop inside another.

What happens if the condition in a `for` loop is false at start?

The loop does not execute.

What is an example of a `for` loop?

```
for(int i = 0; i < 10; i++)
```

Can a `for` loop run infinitely?

Yes, if the condition never becomes false.

What is a loop counter?

A variable used to count loop iterations.

Lecture 10

What is a nested loop?

A loop inside another loop.

When are nested loops commonly used?

For processing matrices or tables.

How does a nested `for` loop work?

The inner loop runs completely for each outer loop iteration.

How many total iterations in a 2-level nested loop?

Multiply outer loop count by inner loop count.

Can you use different types of loops in nesting?

Yes, any combination of loops can be nested.

How do you control nesting levels?

Proper indentation and understanding logic.

Why avoid deep nesting?

It makes code complex and harder to debug.

What can go wrong in nested loops?

Wrong loop bounds, missed conditions, or logic errors.

What is a good practice in nested loops?

Use meaningful variable names and avoid too many levels.

How do you print a pattern using nested loops?

Outer loop for rows, inner loop for columns.

Lecture 11

What are functions in C?

Blocks of code that perform specific tasks.

Why use functions in programming?

To avoid repetition and organize code better.

What is the syntax of a function?

```
returnType functionName(parameters) { statements }
```

What is a function prototype?

A declaration that tells the compiler about the function.

What is a function call?

Invoking the function to execute its code.

What are function parameters?

Variables passed to the function for use.

What is a return value?

The result a function gives back after execution.

What is the `main()` function?

The entry point of every C program.

How do you define a void function?

With `void` return type and no return value.

What is modular programming?

Breaking code into smaller, reusable functions.

Lecture 12

What is recursion?

A function calling itself.

When is recursion useful?

For problems that can be broken into smaller similar problems.

What is the base case in recursion?

A condition that stops the recursion.

What is an example of recursion?

Factorial: `fact(n) = n * fact(n-1)`

What happens if the base case is missing?

It leads to infinite recursion and a crash.

What is a recursive function?

A function that solves a problem by calling itself.

How does the system handle recursive calls?

Using the call stack to store return points.

What is a drawback of recursion?

Consumes more memory and is slower than loops.

How do you convert recursion to iteration?

By using loops and explicit stacks.

Why study recursion?

It helps understand problem-solving and function flow.

Lecture 13

What is an array?

A collection of elements of the same type.

How do you declare an array in C?

```
dataType arrayName[size];
```

What is the index of the first array element?

Zero (0).

How do you access an array element?

Using its index, like `arr[2]`.

What happens if you access out-of-bound index?

Undefined behavior or runtime error.

What is the purpose of arrays?

To store and manage multiple values efficiently.

Can you use loops with arrays?

Yes, loops are ideal for processing arrays.

What is the maximum size of an array?

Depends on system memory and declared size.

How do you input array values?

Using a loop with `cin >> arr[i];`

Can arrays be passed to functions?

Yes, by passing the array name (pointer).

Lecture 14

What is a string in C?

An array of characters ending with `'\0'`.

How do you declare a string?

```
char name[20];
```

What is the difference between character and string?

A character is a single letter; a string is multiple characters.

What is the null character?

`'\0'`, marks the end of a string.

How do you input a string in C?

Using `cin >>` or `gets()`.

How do you display a string?

Using `cout <<` or `puts()`.

Why avoid `gets()` in modern C++?

It causes buffer overflows; `cin.getline()` is safer.

What header file is used for string functions?

```
#include <cstring>.
```

What is `strlen()` used for?

To get the length of a string.

What does `strcpy()` do?

Copies one string into another.

Lecture 15

What is the purpose of `strcmp()`?

To compare two strings.

What does `strcat()` do?

Concatenates two strings.

What does `strrev()` do?

Reverses a string.

What does `strupr()` do?

Converts a string to uppercase.

What does `strlwr()` do?

Converts a string to lowercase.

What does `strncmp()` do?

Compares first `n` characters of two strings.

Can you manipulate strings using loops?

Yes, by accessing each character via index.

What is a limitation of strings in C?

Fixed size and manual memory handling.

What is a character array?

An array used to store a string.

What happens if a string is not null-terminated?

Functions like `strlen()` will not work correctly.

Lecture 16

What is the difference between local and global variables?

Local variables are declared inside functions; global outside.

Where is a local variable accessible?

Only within the block it is declared.

Where is a global variable accessible?

Throughout the program.

What is a scope of a variable?

The region where the variable is valid.

What is variable shadowing?

A local variable with the same name hides a global one.

Why should global variables be avoided?

They reduce modularity and can cause conflicts.

What is the lifetime of a local variable?

From declaration till the end of the block.

What is the lifetime of a global variable?

Entire duration of program execution.

How do you declare a global variable?

Outside all functions, typically at the top.

What is the `static` keyword used for?

To retain the variable value between function calls.

Lecture 17

What is a constant in C?

A value that cannot be changed during program execution.

How do you declare a constant in C?

Using `const`, e.g., `const int x = 10;`

Why use constants?

To prevent accidental changes to values that should remain fixed.

What is a macro?

A constant defined using `#define`, e.g., `#define PI 3.14`

What is the difference between `const` and `#define`?

`const` has type checking; `#define` is a preprocessor directive.

When should you use `#define`?

For symbolic constants and macros.

Can you use `const` in function parameters?

Yes, to prevent modification of passed arguments.

What is the scope of a constant variable?

Same as regular variables—depends on where it's declared.

Can `const` be used with pointers?

Yes, to make pointer or pointee read-only.

Why prefer `const` over `#define`?

Because it respects data types and scopes.

Lecture 18

What are storage classes in C?

They define the scope, lifetime, and visibility of variables.

What are the four storage classes in C?

`auto`, `register`, `static`, `extern`.

What is the default storage class for local variables?

`auto`.

What does the `register` keyword suggest?

Store the variable in CPU register for faster access.

What does `static` do to a local variable?

Preserves its value between function calls.

What is an `extern` variable?

A global variable declared in another file.

What is the lifetime of a `static` variable?

Entire program duration, even if it's local.

When should you use `register`?

For frequently used variables like loop counters.

What does `auto` keyword indicate?

Default for local variables; rarely used explicitly.

How does `extern` help in multi-file programs?

It allows variable sharing across files.

Lecture 19

What is a pointer?

A variable that stores the address of another variable.

How do you declare a pointer?

```
dataType *ptr; e.g., int *p;
```

What is the address-of operator?

`&`, used to get the address of a variable.

What is the dereference operator?

`*`, used to access the value pointed by a pointer.

How do you assign a value to a pointer?

`ptr = &x;` assigns the address of `x` to `ptr`.

What does `*ptr = 5;` do?

Sets the value of the variable pointed by `ptr` to 5.

Can pointers point to different data types?

Yes, but they must match the data type they're pointing to.

What is a null pointer?

A pointer that doesn't point to any valid memory location.

What is pointer arithmetic?

Operations like `ptr++` to move to the next memory location.

Why use pointers?

For efficient memory access, dynamic memory, and arrays.

Lecture 20

What is dynamic memory allocation?

Allocating memory at runtime using pointers.

Which functions are used for dynamic memory in C?

`malloc()`, `calloc()`, `realloc()`, `free()`.

What does `malloc()` do?

Allocates a block of memory and returns a pointer to it.

What is the return type of `malloc()`?

`void*`, usually typecast to the desired pointer type.

What does `free()` do?

Deallocates memory previously allocated with `malloc()` or `calloc()`.

What is `calloc()`?

Allocates multiple blocks and initializes them to zero.

What is the use of `realloc()`?

Resizes previously allocated memory.

Why is memory deallocation important?

To avoid memory leaks and wastage.

What happens if you access freed memory?

It leads to undefined behavior.

How do you allocate memory for an array dynamically?

Using `malloc(size * sizeof(type))`.

Lecture 21

What is a structure in C?

A user-defined data type that groups different variables.

How do you define a structure?

Using `struct` keyword.

What is the syntax to declare a structure?

```
struct student { int id; char name[20]; };
```

How do you declare a structure variable?

```
struct student s1;
```

How do you access members of a structure?

Using dot operator, e.g., `s1.id`

Can structures contain different data types?

Yes, unlike arrays which must be the same type.

What is a nested structure?

A structure having another structure as a member.

How do you pass a structure to a function?

By value or by reference (using pointers).

Can you assign one structure to another?

Yes, if both are of the same type.

Why use structures?

To manage related data in a single unit.

Lecture 22

What is a union in C?

A data type where all members share the same memory.

How is a union declared?

Using `union` keyword, similar to `struct`.

What is the size of a union?

Equal to the size of its largest member.

How do you access union members?

Using dot operator, e.g., `u1.a`

Can all union members be used at the same time?

No, only one member holds a valid value at a time.

What is the difference between `struct` and `union`?

Structures store all members; unions share memory.

Why use a union?

To save memory when only one member is used at a time.

Can unions be nested like structures?

Yes, unions can contain structures and vice versa.

Can you pass a union to a function?

Yes, like structures—by value or by pointer.

Where are unions useful?

In applications where memory is limited or data types vary.

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